

베이지안 기계학습

4. Bayesian inference for regression models I

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Bayesian inference for linear regression models

Linear regression models

$$Y_i \mid \beta, \sigma^2 \stackrel{\text{ind}}{\sim} N(X_i^T \beta, \sigma^2), \quad i = 1, \dots, n$$

- ▶ $Y_i \in \mathbb{R}$, $X_i \in \mathbb{R}^p$, $\beta = (\beta_1, \dots, \beta_p)^T \in \mathbb{R}^p$ and $\sigma^2 > 0$.
- ▶ Let $Y = (Y_1, \dots, Y_n)^T \in \mathbb{R}^n$ and $X = (X_1, \dots, X_n)^T \in \mathbb{R}^{n \times p}$.
- ▶ We have two parameters: (β, σ^2) .

Linear regression models

$$Y_i \mid \beta, \sigma^2 \stackrel{\text{ind}}{\sim} N(X_i^T \beta, \sigma^2), \quad i = 1, \dots, n$$

- ▶ For Bayesian inference, we need to specify priors for (β, σ^2) .
- ▶ While there are various available priors, in this lecture, we focus on
 - the Zellner's *g*-prior for β and
 - the Jeffreys' prior for σ^2 .
- ▶ Priors will be specified in the form $\pi(\beta, \sigma^2) = \pi(\beta \mid \sigma^2)\pi(\sigma^2)$.

Zellner's g -prior

$$\beta \mid \sigma^2 \sim N_p\left(0, g \sigma^2 (X^T X)^{-1}\right), \quad (\text{Zellner's } g\text{-prior})$$

$$\pi(\sigma^2) \propto \frac{1}{\sigma^2}. \quad (\text{Jeffreys' prior})$$

- ▶ The Zellner's g -prior has been commonly used prior in linear regression models.
- ▶ It is a multivariate normal distribution with a specially designed covariance matrix for computational convenience.
- ▶ The prior variance of β must include σ^2 for conjugacy.

*Posterior

- ▶ We can derive the joint posterior

$$\begin{aligned}\beta \mid \sigma^2, y &\sim N_p\left(\frac{g}{g+1}\hat{\beta}, \frac{\sigma^2 g}{g+1}(X^T X)^{-1}\right), \\ \sigma^2 \mid y &\sim IG\left(\frac{n}{2}, \frac{n\hat{\sigma}^2}{2} + \frac{1}{2(g+1)}\hat{\beta}^T X^T X \hat{\beta}\right),\end{aligned}$$

where $\hat{\beta} = (X^T X)^{-1} X^T y$ and $\hat{\sigma}^2 = \|y - X\hat{\beta}\|_2^2/n$.

- ▶ In this case, we don't need an MCMC algorithm for posterior inference.

*Posterior computation

(Full conditional posterior of β)

$$\begin{aligned}\pi(\beta \mid \sigma^2, y) &\propto f(y \mid \beta, \sigma^2) \pi(\beta \mid \sigma^2) \\ &\propto \exp \left(-\frac{1}{2\sigma^2} \|y - X\beta\|_2^2 - \frac{1}{2g\sigma^2} \beta^T X^T X \beta \right) \\ &\propto \exp \left\{ -\frac{1}{2} \left(\beta^T \frac{g+1}{g\sigma^2} X^T X \beta - 2\beta^T \frac{1}{\sigma^2} X^T y \right) \right\} \\ &\stackrel{(*)}{\propto} \text{pdf of } N_p \left(\frac{g}{g+1} \hat{\beta}, \frac{\sigma^2 g}{g+1} (X^T X)^{-1} \right)\end{aligned}$$

(*) Note that $N_p(\beta \mid \mu, \Sigma) \propto \exp \left\{ -\frac{1}{2} (\beta^T \Sigma^{-1} \beta - 2\beta^T \Sigma^{-1} \mu) \right\}$.

*Posterior computation

(Marginal posterior of σ^2)

$$\begin{aligned}\pi(\sigma^2 | y) &\propto \int f(y | \beta, \sigma^2) \pi(\beta | \sigma^2) \pi(\sigma^2) d\beta \\ &\propto \int (\sigma^2)^{-\frac{n+p}{2}-1} \exp\left(-\frac{1}{2\sigma^2} \|y - X\beta\|_2^2 - \frac{1}{2g\sigma^2} \beta^T X^T X \beta\right) d\beta \\ &\propto \int (\sigma^2)^{-\frac{n+p}{2}-1} \exp\left(-\frac{1}{2\sigma^2} y^T y\right) \\ &\quad \times \exp\left\{-\frac{1}{2}\left(\beta^T \frac{g+1}{g\sigma^2} X^T X \beta - 2\beta^T \frac{1}{\sigma^2} X^T y\right)\right\} d\beta \\ &\propto (\sigma^2)^{-\frac{n}{2}-1} \exp\left\{-\frac{1}{2\sigma^2} y^T y + \frac{g}{2\sigma^2(g+1)} y^T X (X^T X)^{-1} X^T y\right\} \\ &\quad \times \underbrace{\int \text{pdf of } N_p\left(\frac{g}{g+1} \hat{\beta}, \frac{\sigma^2 g}{g+1} (X^T X)^{-1}\right) d\beta}_{=1},\end{aligned}$$

*Posterior computation

where

$$\begin{aligned}y^T y - \frac{g}{g+1} y^T X (X^T X)^{-1} X^T y &= y^T y - \frac{g}{g+1} \hat{\beta}^T X^T X \hat{\beta} \\ &= \|y - X \hat{\beta}\|_2^2 + \frac{1}{g+1} \hat{\beta}^T X^T X \hat{\beta}.\end{aligned}$$

Therefore, we have

$$\pi(\sigma^2 | y) \propto (\sigma^2)^{-\frac{n}{2}-1} \exp \left[-\frac{1}{\sigma^2} \left\{ \frac{\|y - X \hat{\beta}\|_2^2}{2} + \frac{1}{2(g+1)} \hat{\beta}^T X^T X \hat{\beta} \right\} \right],$$

which implies

$$\sigma^2 | y \sim IG\left(\frac{n}{2}, \frac{n\hat{\sigma}^2}{2} + \frac{1}{2(g+1)} \hat{\beta}^T X^T X \hat{\beta}\right).$$

Remark: choice of g

- ▶ (Fixed g) Note that

$$\mathbb{E}(\beta \mid \sigma^2, y) = \frac{1}{g+1} \underbrace{0}_{(\text{prior})} + \frac{g}{g+1} \underbrace{\hat{\beta}}_{(\text{data})}.$$

For example, $g = n$ means prior has the equivalent weight of 1 observation.

- ▶ (Fully Bayesian) We can use

$$g \sim IG(1/2, n/2). \quad (\text{Zellner-Siow prior})^1$$

¹Zellner and Siow (1980) Posterior odds ratios for selected regression hypotheses.

Example

$$\begin{aligned}Y_i \mid \beta, \sigma^2 &\stackrel{\text{ind}}{\sim} N(X_i^T \beta, \sigma^2), \quad i = 1, \dots, n, \\ \beta \mid \sigma^2 &\sim N_p\left(0, g\sigma^2(X^T X)^{-1}\right), \\ \pi(\sigma^2) &\propto 1/\sigma^2, \\ g &\sim \text{IG}(1/2, n/2)\end{aligned}$$

- ▶ Consider the above linear regression model.
- ▶ We will use the **Zellner-Siow prior**.
- ▶ Note that the joint posterior is no longer available due to the prior on the hyperparameter g .

R code

```
library(rstan)

# data generation
n <- 1000
p <- 5
X <- matrix(rnorm(n*p), nrow=n, ncol=p)
beta.true <- c(1,1,1,2,2)
sig.true <- 1

set.seed(1)
y <- rnorm(n, X %*% beta.true, sd = sig.true)
```

R code

```
# stan code
reg.model <- "
  data{
    int<lower=0> n;
    int<lower=0> p;
    real y[n];
    matrix[n,p] X;
  }
  transformed data{
    vector[p] zerovec;
    zerovec = rep_vector(0., p);
    matrix[p,p] inv_XtX;
    inv_XtX = inverse(X' * X);
  }
```

R code

```
parameters{
  vector[p] beta;
  real<lower=0> g;
  real<lower=0> sigma2;
}
model{
  // likelihood
  y ~ normal(X*beta, sqrt(sigma2));
  // priors
  g ~ inv_gamma(0.5, n*0.5);
  beta ~ multi_normal(zerovec, g*sigma2*inv_XtX);
  target += -log(sigma2);
}
```

"

R code

```
# compile the model
stanC <- stan_model(model_code = reg.model)

# data
stanD <- list(y = y, X = X, n = n, p = p)

# posterior inference
stanF <- sampling(object = stanC, data = stanD,
                  chain = 3, iter = 1000,
                  warmup = 200, thin = 1)
```

R code

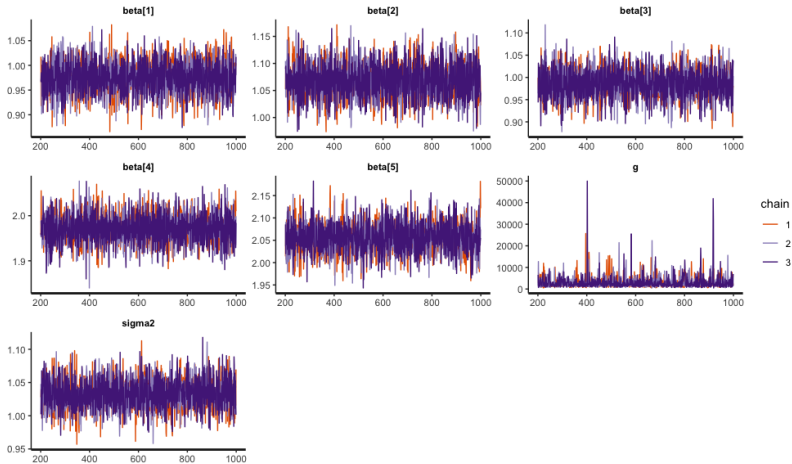
```
print(stanF)
```

```
> print(stanF)
Inference for Stan model: 17827444c7d7877364447d86dd5e3dc6.
3 chains, each with iter=1000; warmup=200; thin=1;
post-warmup draws per chain=800, total post-warmup draws=2400.
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
beta[1]	0.98	0.00	0.03	0.91	0.95	0.98	1.00	1.04	3741	1.00
beta[2]	1.07	0.00	0.03	1.01	1.05	1.07	1.09	1.13	3662	1.00
beta[3]	0.99	0.00	0.03	0.92	0.96	0.99	1.01	1.05	3307	1.00
beta[4]	1.97	0.00	0.03	1.91	1.95	1.97	1.99	2.03	4157	1.00
beta[5]	2.06	0.00	0.03	1.99	2.03	2.06	2.08	2.12	3096	1.00
g	2976.69	68.73	2747.74	813.99	1521.52	2238.32	3419.25	9364.63	1598	1.00
sigma2	1.03	0.00	0.02	0.99	1.02	1.03	1.05	1.08	3468	1.00
lp__	-541.33	0.05	1.87	-545.69	-542.41	-540.98	-539.93	-538.70	1272	1.01

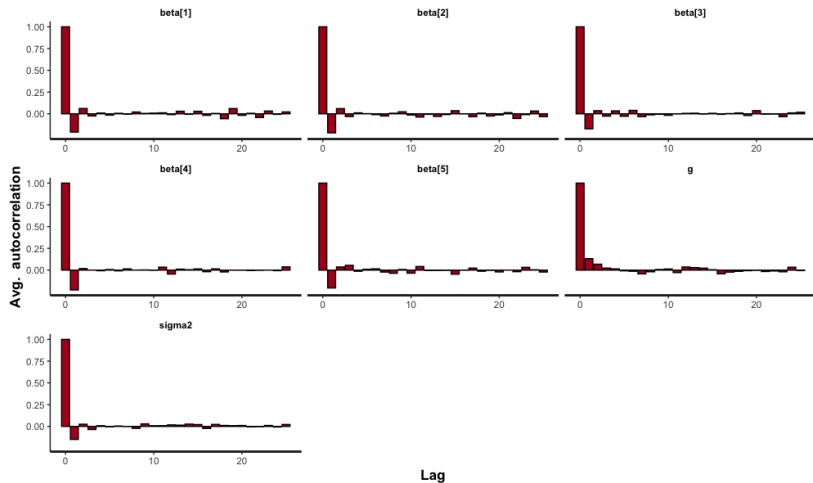
R code

```
stan_trace(stanF)
```



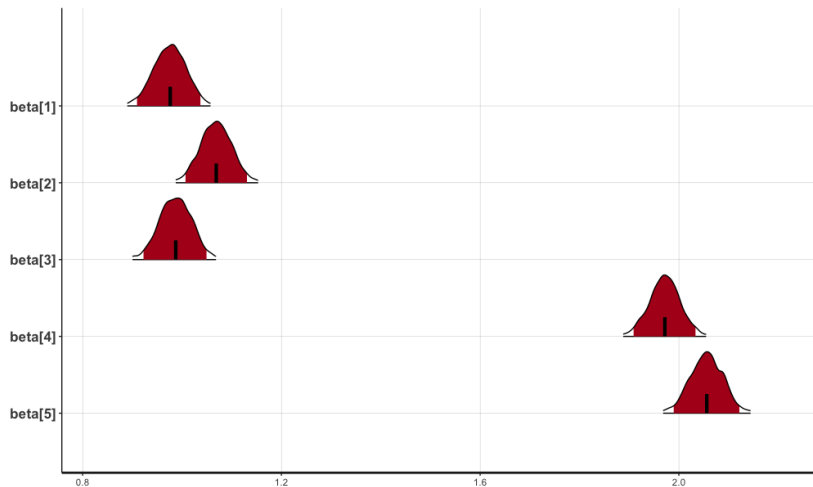
R code

```
stan_ac(stanF)
```



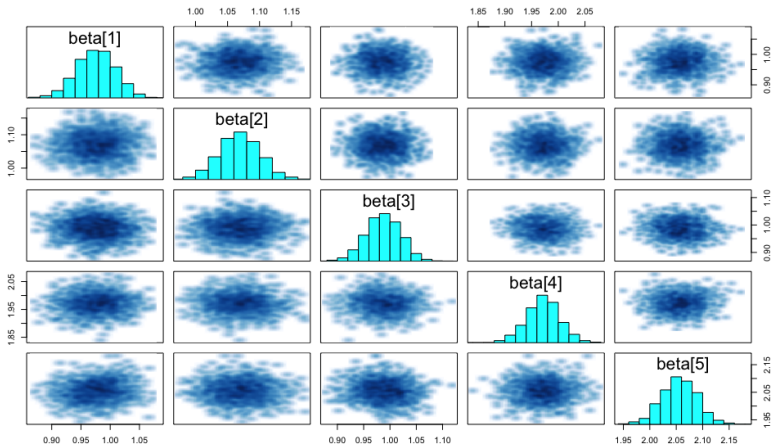
R code

```
plot(stanF, show_density = T, ci_level = 0.95,  
     pars = "beta", outer_level = 0.99)
```



R code

```
pairs(stanF, pars = "beta")
```



Prediction

- ▶ In this model, let's say we want to predict the Y_i values for new X_i values.
- ▶ We can perform the prediction with rstan package by using `generated_quantities` block.

R code: prediction

```
reg.model.pred <- "  
  data{  
    int<lower=0> n;  
    int<lower=0> p;  
    vector[n] y;  
    matrix[n,p] X;  
    int<lower=0> n_new;  
    matrix[n_new,p] X_new;  
  }  
  transformed data{  
    vector[p] zerovec;  
    zerovec = rep_vector(0., p);  
    matrix[p,p] inv_XtX;  
    inv_XtX = inverse(X' * X);  
  }
```


R code: prediction

```
parameters{
  vector[p] beta;
  real<lower=0> g;
  real<lower=0> sigma2;
}
model{
  // likelihood
  y ~ normal(X*beta, sqrt(sigma2));
  // priors
  g ~ inv_gamma(0.5, n*0.5);
  beta ~ multi_normal(zerovec, g*sigma2*inv_XtX);
  target += -log(sigma2);
}
```

R code: prediction

```
generated quantities{  
  vector[n_new] y_new;  
  for(i in 1:n_new){  
    y_new[i] = normal_rng(X_new[i,]*beta,  
                          sqrt(sigma2));  
  }  
}
```

"

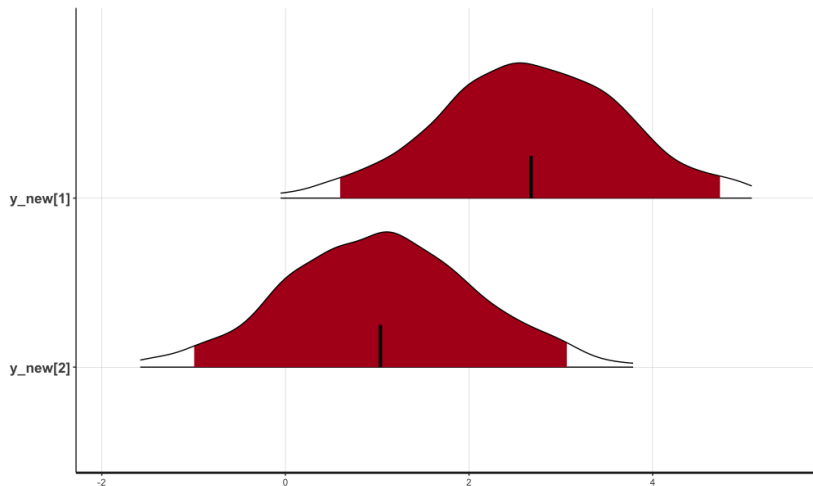
R code: prediction

```
print(stanF)
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
beta[1]	0.98	0.00	0.03	0.91	0.95	0.98	1.00	1.04	3608	1
beta[2]	1.07	0.00	0.03	1.00	1.05	1.07	1.09	1.13	3979	1
beta[3]	0.99	0.00	0.03	0.92	0.97	0.99	1.01	1.05	4108	1
beta[4]	1.97	0.00	0.03	1.91	1.95	1.97	1.99	2.03	4454	1
beta[5]	2.06	0.00	0.03	1.99	2.03	2.06	2.08	2.12	4218	1
g	2850.64	68.60	2501.22	796.42	1499.64	2180.27	3276.82	9576.54	1330	1
sigma2	1.07	0.00	0.05	0.98	1.04	1.07	1.10	1.16	3523	1
y_new[1]	2.67	0.02	1.03	0.60	1.98	2.67	3.37	4.73	2482	1
y_new[2]	1.03	0.02	1.04	-0.99	0.32	1.03	1.73	3.07	2319	1
lp__	-541.32	0.06	1.89	-545.79	-542.39	-541.03	-539.92	-538.72	1104	1

R code: prediction

```
plot(stanF, show_density = T, ci_level = 0.95,  
     pars = "y_new", outer_level = 0.99)
```



Example: immigrants data²

$$Y_i \mid \beta, \sigma^2 \stackrel{ind}{\sim} N(X_i^T \beta, \sigma^2), \quad i = 1, \dots, n = 35.$$

- ▶ Y_i : average wage of immigrants from the i th country.
- ▶ $X_i = (1, z_{1i}, z_{2i})$, where
 - z_{1i} : percentage of English speakers,
 - z_{2i} : percentage of people who can read.

²오만숙 (2019) JAGS를 활용한 베이지안 자료분석.

R code

```
// read data
im.dat = read.csv("immigrants.csv")

y = im.dat$wage
n = nrow(im.dat)

X = matrix(1, nrow=n, ncol=3)
X[,2] = im.dat$sp
X[,3] = im.dat$lit
p = ncol(X)

X[,2:3] = scale(X[,2:3])
```

R code

```
// posterior inference
library(rstan)

// use the "stanC" object in the previous example
stanD <- list(y = y, X = X, n = n, p = p)
stanF <- sampling(object = stanC, data = stanD,
                  chain = 3, iter = 1000,
                  warmup = 200, thin = 1)
```

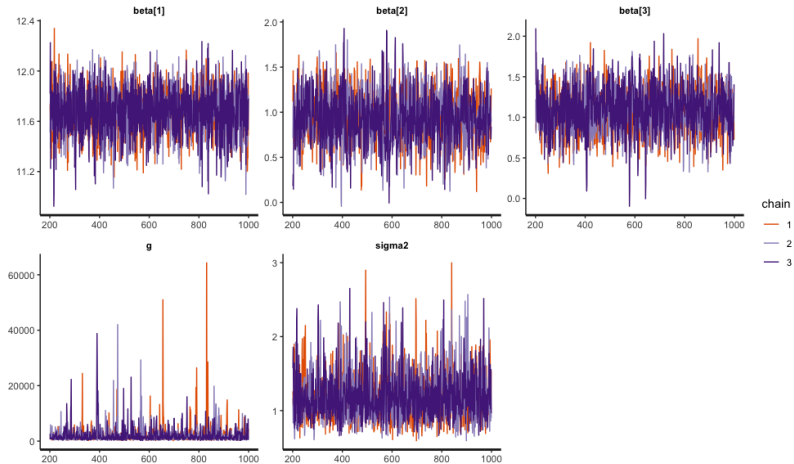
R code

```
print(stanF)
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
beta[1]	11.67	0.00	0.19	11.29	11.55	11.67	11.80	12.01	2877	1
beta[2]	0.94	0.01	0.27	0.38	0.77	0.94	1.12	1.48	1658	1
beta[3]	1.08	0.01	0.27	0.54	0.89	1.08	1.26	1.61	1686	1
g	2120.97	94.35	3211.62	342.53	759.62	1302.86	2341.47	9012.30	1159	1
sigma2	1.19	0.01	0.32	0.72	0.97	1.15	1.36	1.94	1795	1
lp__	-31.31	0.05	1.66	-35.54	-32.13	-30.94	-30.11	-29.14	1121	1

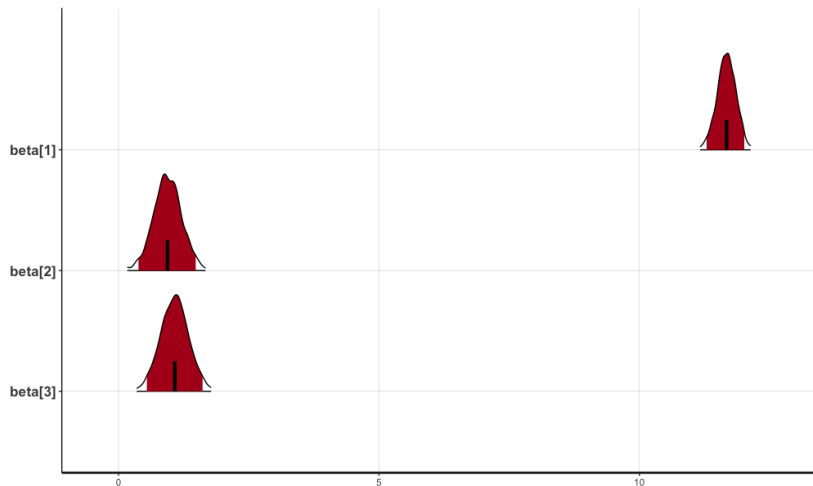
R code

```
stan_trace(stanF)
```



R code

```
plot(stanF, show_density = T, ci_level = 0.95,  
     pars = "beta", outer_level = 0.99)
```



R code

```
pairs(stanF, pars = "beta")
```

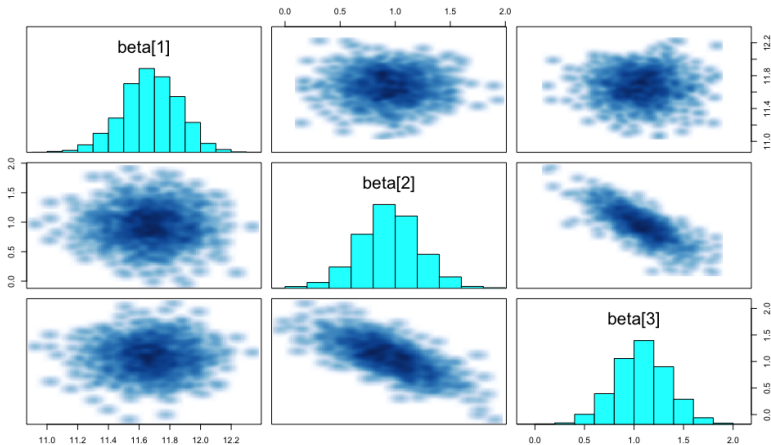


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Bayesian inference for generalized linear regression models

Logistic regression

$$\begin{aligned} Y_i \mid \beta &\overset{ind}{\sim} \text{Ber}(p_i), \\ \log\left(\frac{p_i}{1-p_i}\right) &= \alpha + X_i^T \beta, \quad i = 1, \dots, n. \end{aligned}$$

- ▶ $Y_i \in \{0, 1\}$: binary response variable
- ▶ $X_i \in \mathbb{R}^p$: covariate vector
- ▶ Basically, we can apply the priors we used for β in the linear regression model here as well.

Example: Default data

$$\begin{aligned}Y_i | \beta &\overset{\text{ind}}{\sim} \text{Ber}(p_i), \\ \log\left(\frac{p_i}{1-p_i}\right) &= \alpha + X_i^T \beta, \quad i = 1, \dots, n = 10000, \\ \alpha &\sim N(0, 100), \\ \beta | g &\sim N_p\left(0, g(X^T X)^{-1}\right), \\ \pi(\sigma^2) &\propto 1/\sigma^2, \\ g &\sim IG(1/2, n/2)\end{aligned}$$

- ▶ Y_i : i th individual's bankruptcy status
- ▶ $X_i = (X_{i1}, X_{i2}, X_{i3})^T$ ($p = 3$), where
 - student status,
 - credit card debt balance,
 - annual income.

R code³

```
library(rstan)
library(ISLR) # to use Default data

dim(Default)
head(Default)
```

```
> dim(Default)
[1] 10000      4
> head(Default)
  default student  balance  income
1      No      No  729.5265 44361.625
2      No     Yes  817.1804 12106.135
3      No      No 1073.5492 31767.139
4      No      No  529.2506 35704.494
5      No      No  785.6559 38463.496
6      No     Yes  919.5885  7491.559
```

³이재용, 이기재 (2022) 베이지 데이터 분석. 한국방송통신대학교출판문화원.

R code

```
y = as.numeric(Default$default) - 1
n = nrow(Default)

X = matrix(1, nrow=n, ncol=3)
X[,1] = as.numeric(Default$student) - 1
X[,2] = Default$balance
X[,3] = Default$income
X[,2:3] = scale(X[,2:3])
p = ncol(X)
```

R code

```
logistic.model <- "  
  data{  
    int<lower=0> n;  
    int<lower=0> p;  
    int<lower=0, upper=1> y[n];  
    matrix[n,p] X;  
  }  
  transformed data{  
    vector[p] zerovec;  
    zerovec = rep_vector(0., p);  
    matrix[p,p] inv_XtX;  
    inv_XtX = inverse(X' * X);  
  }
```

R code

```
parameters{
  real alpha;
  vector[p] beta;
  real<lower=0> g;
}
model{
  // likelihood
  y ~ bernoulli_logit_glm(X, alpha, beta);
  // priors
  g ~ inv_gamma(0.5, n*0.5);
  beta ~ multi_normal(zerovec, g*inv_XtX);
  alpha ~ normal(0, 10);
}
```

"

R code

```
stanC <- stan_model(model_code = logistic.model)
stanD <- list(y = y, X = X, n = n, p = p)
stanF <- sampling(object = stanC, data = stanD,
  chain = 3, iter = 1000,
  warmup = 200, thin = 1)
```

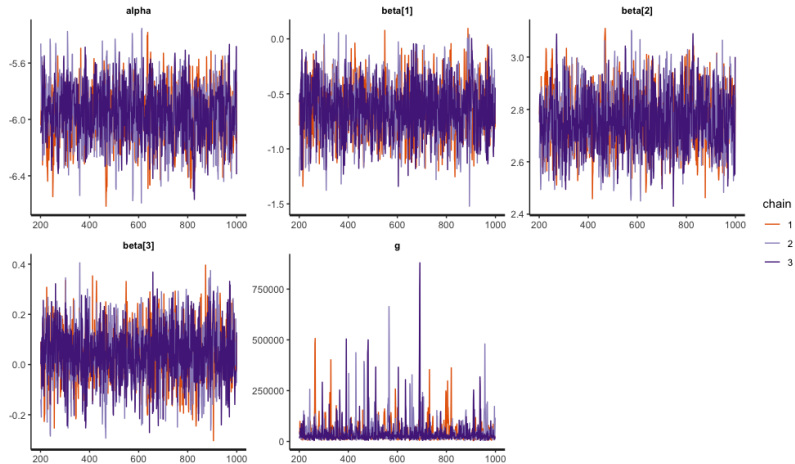
R code

```
print(stanF)
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
alpha	-5.97	0.01	0.19	-6.34	-6.10	-5.97	-5.84	-5.60	1170	1
beta[1]	-0.64	0.01	0.23	-1.11	-0.80	-0.63	-0.47	-0.20	1362	1
beta[2]	2.76	0.00	0.11	2.55	2.69	2.76	2.84	2.98	1175	1
beta[3]	0.04	0.00	0.11	-0.17	-0.03	0.04	0.12	0.25	1443	1
g	41011.07	1931.28	65198.90	7516.03	15505.12	24655.64	42925.38	175752.14	1140	1
lp__	-797.43	0.05	1.58	-801.41	-798.24	-797.13	-796.28	-795.33	970	1

R code

```
stan_trace(stanF)
```



R code

```
plot(stanF, show_density = T, ci_level = 0.95,  
     pars = c("alpha", "beta"), outer_level = 0.99)
```

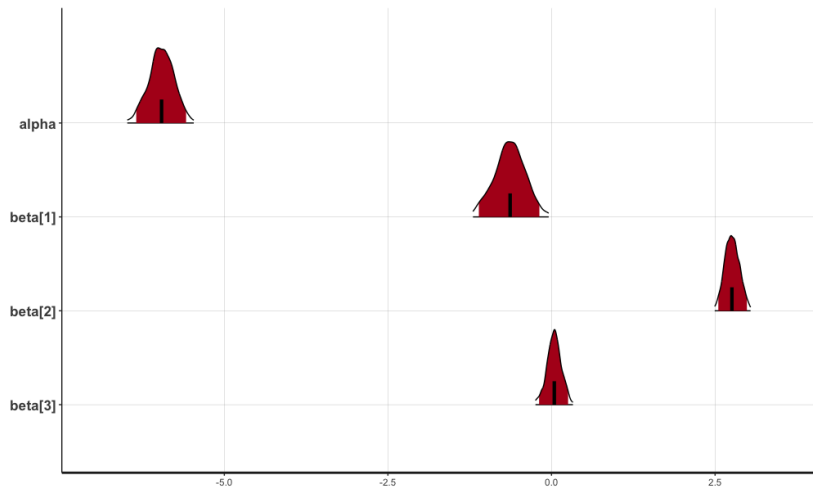


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Remark and next week

Remark

- ▶ The Zellner's g -prior is one of the most popular priors in regression models.
- ▶ However, this prior cannot be used in high-dimensional situations where $p > n$: $(X^T X)^{-1}$ does not exist, making the declaration of the prior problematic.
- ▶ Therefore, we need other priors tailored for high-dimensional settings.

Next week

Bayesian inference for regression models II